

# Document 5: The Inter-Service Message Queue Specification

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## 5.1 Thread Isolation Boundaries

To prevent audio dropouts or GUI sluggishness, the architecture divides tasks among three isolated execution environments. Communication across these boundaries is lock-free.

- **GUI Viewport Thread (Low / UI Priority):** Captures mouse/stylus vector points; renders the 128x128 grid representation at 60Hz.
- **Host Compiler Thread (Medium Worker Priority):** Pops raw frames from the queue; processes pixel chunks and executes routing calculations.
- **Audio DSP Thread (Critical Real-Time Priority):** Processes high-speed block-by-block input/output audio buffers (~128 samples). This thread is non-blocking.

## 5.2 Debounce Optimization Control

- **The Idle Window Rule:** A Pen-Up event does not directly post an execution packet. Instead, it places a deferred message on a thread-safe timer scheduled for 300ms in the future.
- **Interruption Logic:** If an interlocking Pen-Down or secondary Pen-Up event occurs before the 300ms timer expires, the pending task is cancelled and reset.
- **Execution Point:** The snapshot message is sent to the Compiler Queue only when the canvas experiences 300ms of absolute idle time.

## 5.3 Deep-Copy Message Payload Mechanics

To prevent data race conditions where the GUI thread might overwrite pixel values while the background Compiler thread is evaluating them, the engine performs an atomic deep copy of the grid data into a fixed 16 KB array message payload containing an execution tracking timestamp and the full frame snapshot copy.